RESEARCH SUMMARY

Songyao Xue | Fall 2027

How can imperfect quantum resources be transformed into reliable computational advantages?

Research focus. My work examines how noise, limited entanglement, protocol exposure, and finite computational budgets can be modeled and reorganized into secure, resource-efficient, and adaptive quantum information systems. My background in Software Engineering provides a computational perspective grounded in algorithms, scientific programming, reproducibility, and implementation cost.

RESEARCH TRAJECTORY

1. Structural quantum security

Analyzed how Grover-based verification structure can enable key recovery, forgery, and repudiation. Contributed to four reusable protocol-inspection points and a strengthened arbitrated signature design using teleportation and strengthened QOTP.

2. Resource-efficient probabilistic decoding

Reinterpreted Huffman lengths as support budgets in a common Hilbert space; combined group ancilla, conditional POVMs, QEC reliability, MAP decisions, and SDP-based resource certificates. The central result is that improved reliability is meaningful only when it survives explicit overhead accounting.

3. Weak-link quantum communication

Developed a no-signaling framework in which weak, heterogeneous entanglement links are treated as task-aware resources. Derived AND/OR first-order corrections, noise-to-visibility mappings, Bayesian weighting, and cross-platform calibration rules for photonic, trapped-ion, and superconducting settings.

4. Learnable quantum networks

Proposed a graph-based weak-link multi-group quantum game framework with game tensors, sign rules, spectral bounds, and differentiable relaxations. Implemented automated search, repeated trials, topology/noise comparisons, and backend-side circuit checks while separating theorem-level results from surrogate and proxy quantities.

SELECTED METHODS AND PREPARATION

- Quantum protocol and information-flow analysis; graph-based weak-link modeling; first-order and spectral analysis.
- POVM-based state discrimination, semidefinite programming, resource-normalized criteria, Bayesian and MAP decisions.
- Python scientific computing, Qiskit, PennyLane, automated experiments, sensitivity analysis, and visualization.
- LaTeX manuscript preparation, appendix proofs, figure/table generation, reference management, and reproducible research organization.
- B.Eng. in Software Engineering, academic rank Top 10%; supervisor-led reading group in quantum information, quantum communication, cryptography, and quantum computing.

CURRENT RESEARCH OUTPUTS

- On Grover-Based Quantum Signature Schemes — second author; revised and resubmitted.
- A Learnable Unified Framework for Weak-Link Multi-Group Quantum Games — first author; under editorial review.
- Weak-Link Entanglement-Assisted No-Signaling Quantum Communication Protocols — first author; manuscript in preparation.
- Error-Corrected Quantum-Huffman-Inspired Probabilistic Source Decoding — first author; manuscript in preparation.

FUTURE RESEARCH AGENDA

Toward a resource-aware Quantum Internet. I aim to study quantum-network protocols that jointly optimize topology, routing, entanglement quality, security, and task performance; learnable graph-structured quantum systems that adapt under calibrated noise; and unified distributed architectures in which communication, computation, cryptography, **and** error correction are evaluated under the same finite-resource budget.

Quantum Information · Quantum Communication · Quantum Cryptography · Quantum Networks · Quantum Machine Intelligence

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